Mthunzelwa Makwanyane
20241063
NET 731 PRACS – Week 5

Week 5 Practical: RBAC, NSG & Storage Security

Role-Based Access Control | Network Security Groups | Azure Storage

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Azure Tenant / Subscription:	Free Trial Subscription
Date Submitted:	2026/04/03

4. Tasks

Task 1 — Role-Based Access Control (3 marks)

The difference between assigning roles at subscription scope and at resource groups is that, at subscription scope the role will affect the users access to the whole entire subscription which is a lot of access. Roles applied on resource scope levels affects that specific resource group and the resources that are deployed inside it unlike the subscription scope that provides users a role to all the resources inside a subscription, including resource groups and the resources inside them.

To test if the role assignments were truly implemented, I logged on azure with the second user's account called Matthew to which I applied the reader role. To test if the role assignment was implemented, I tried to create a Purview Storage account and an azure notification popped up to notify me that this account cannot create this resource in this subscription.

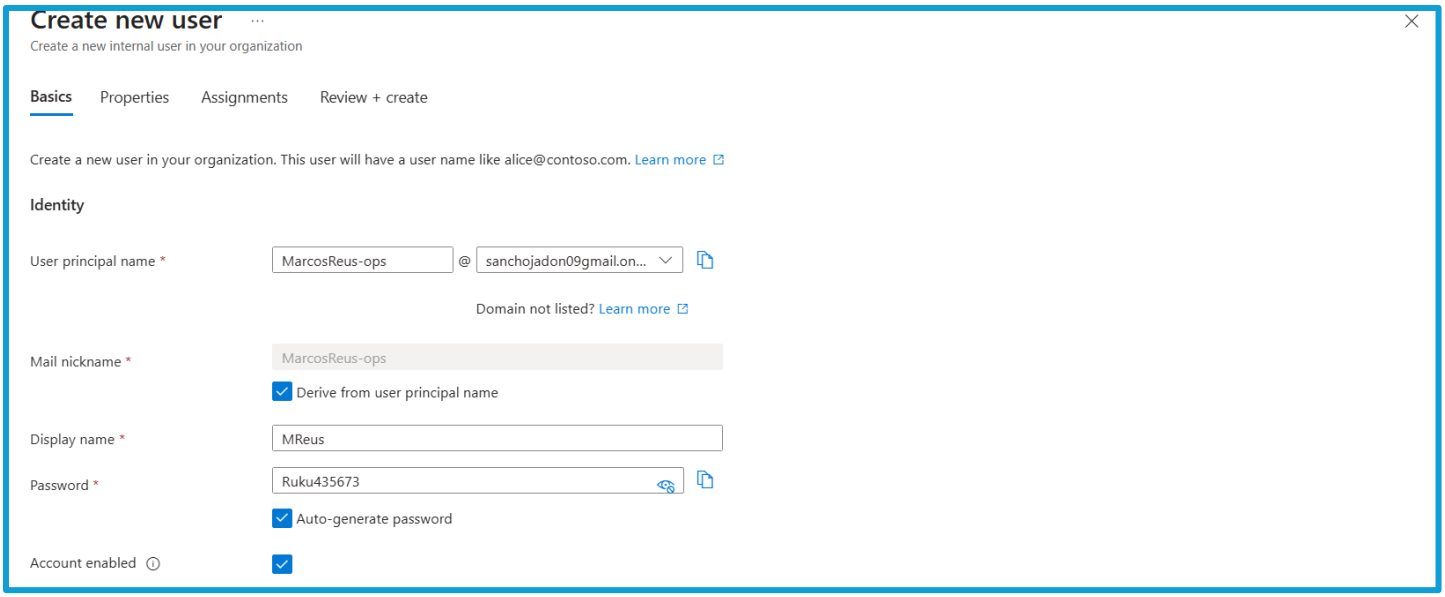

Figure 1: Creating the operations user account

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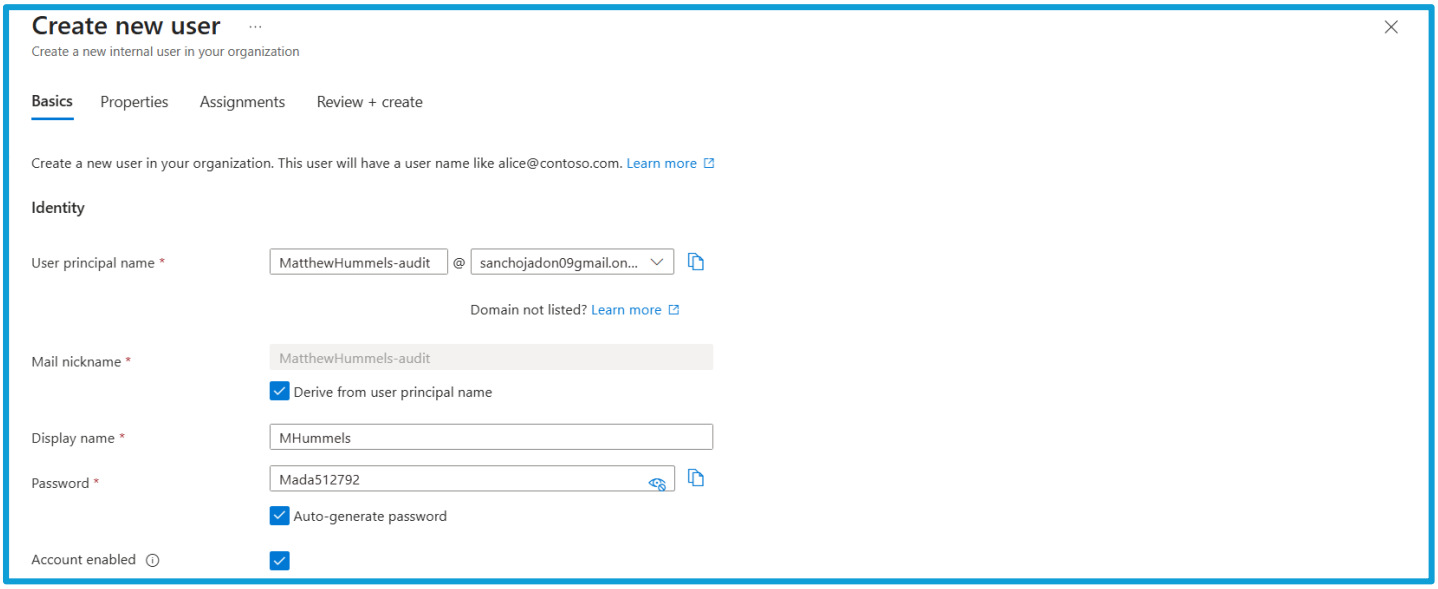

Figure 2: Creating the audit user account

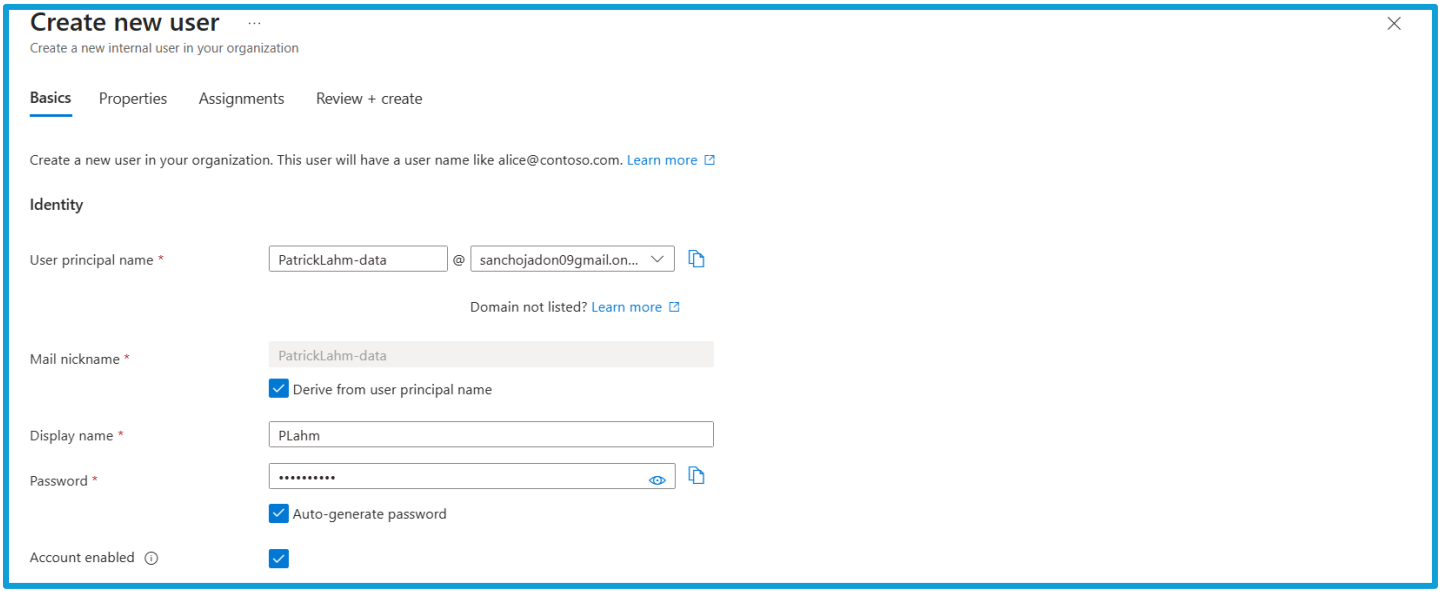

Figure 3: Creating third data user account

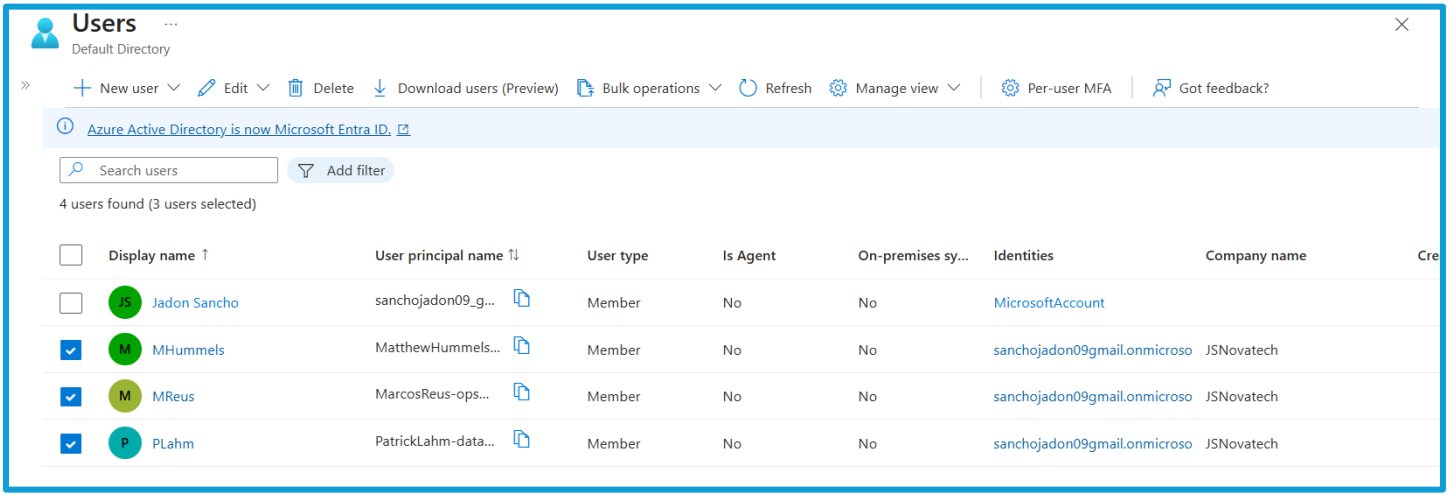

Figure 4: All 3 new created users

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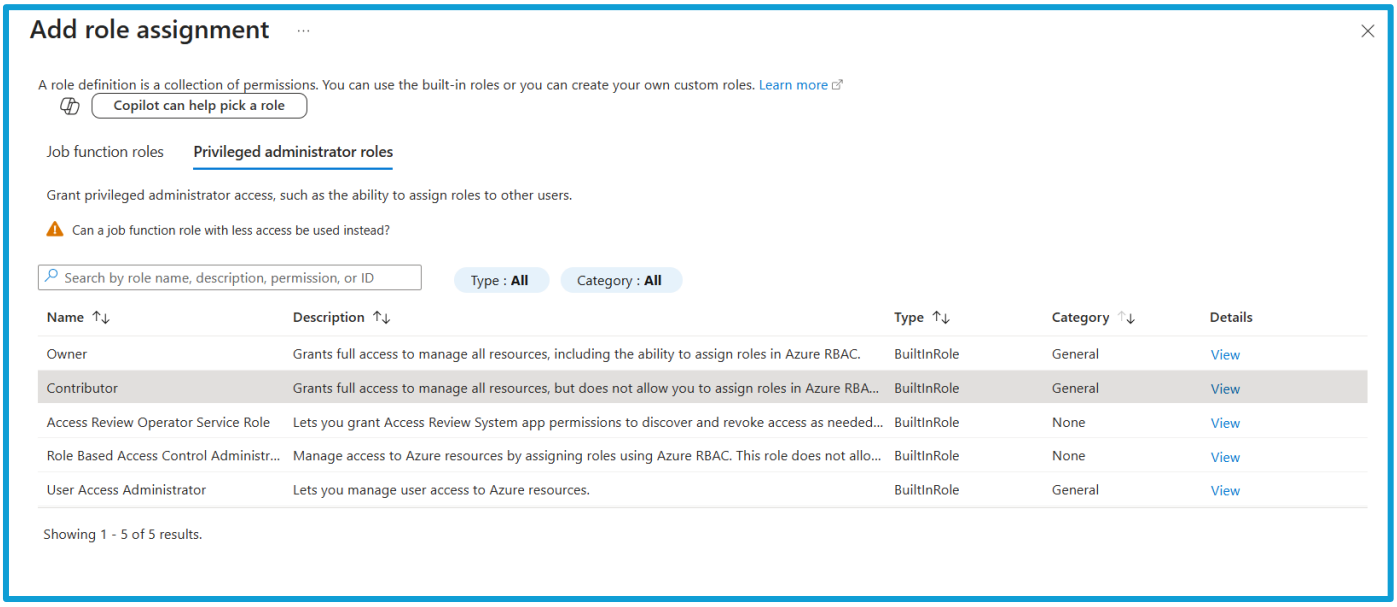

Figure 5: First assignment for the first user

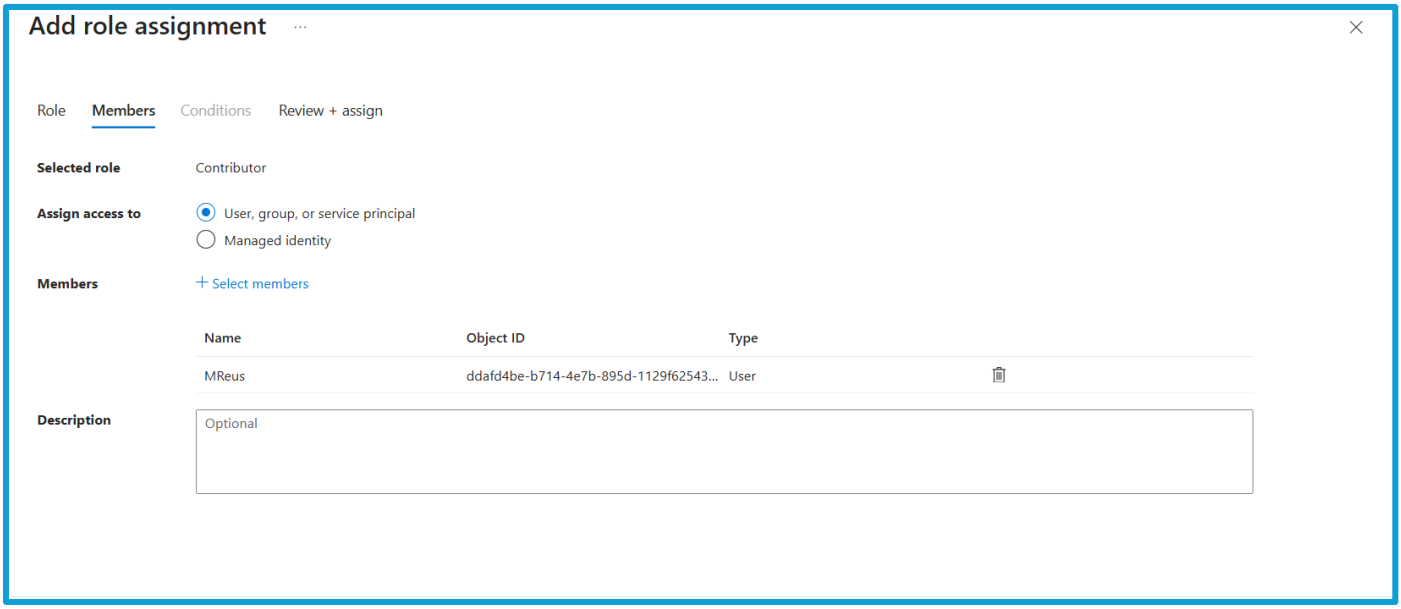

Figure 6: Selecting user for the assignment role

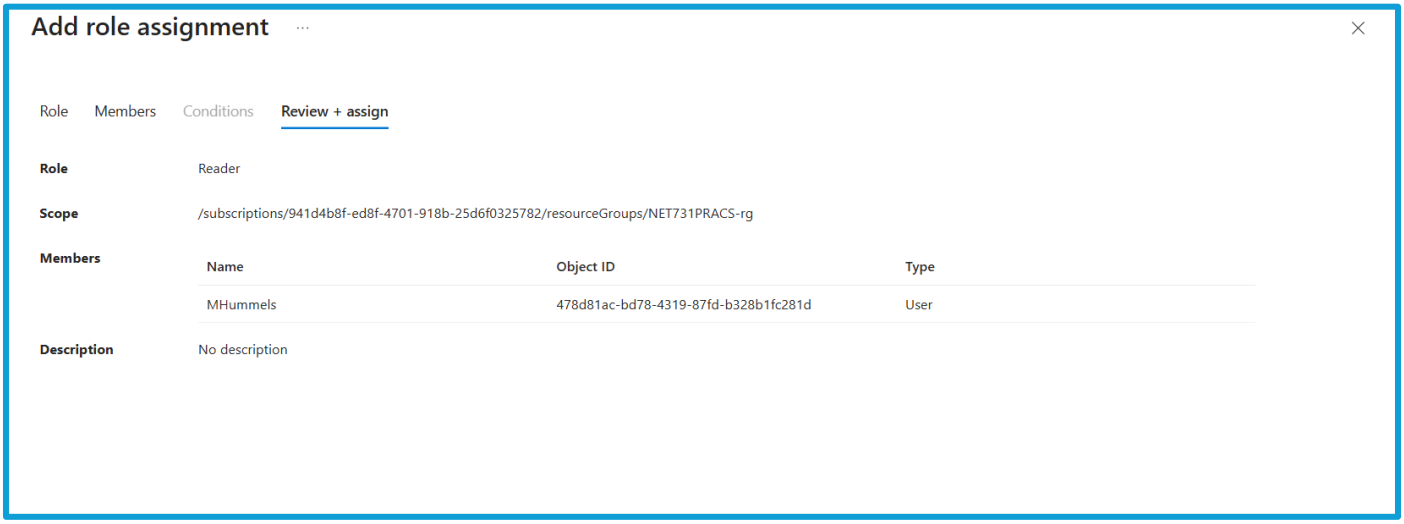

Figure 7: configurations of the second user role for the second user

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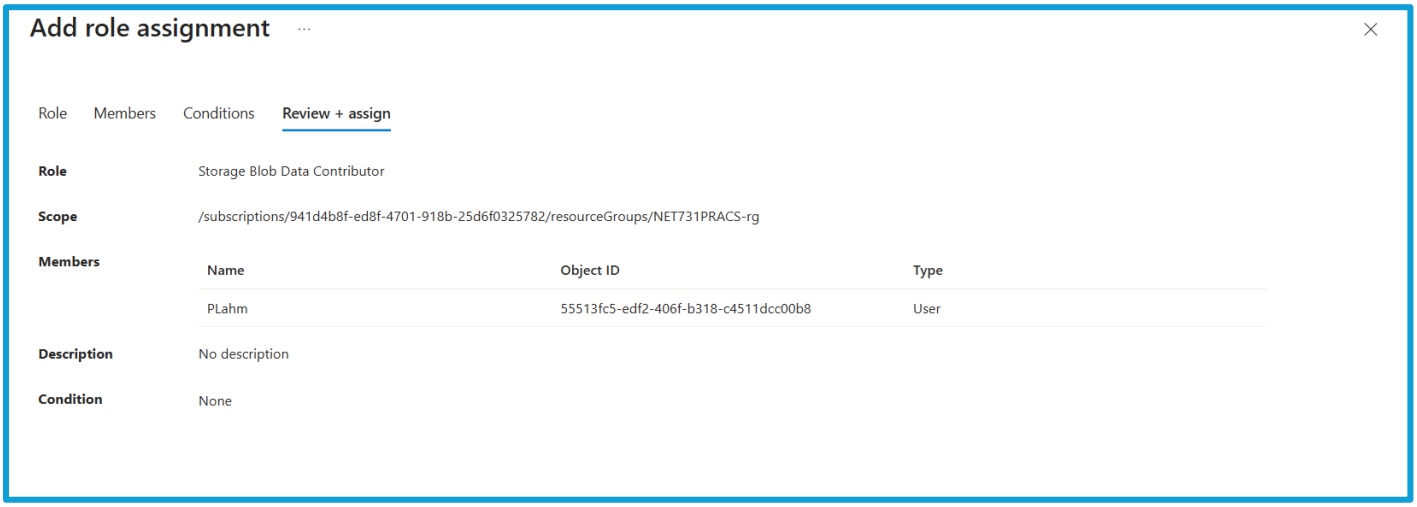

Figure 8: Configurations of the third role assignment for the third user

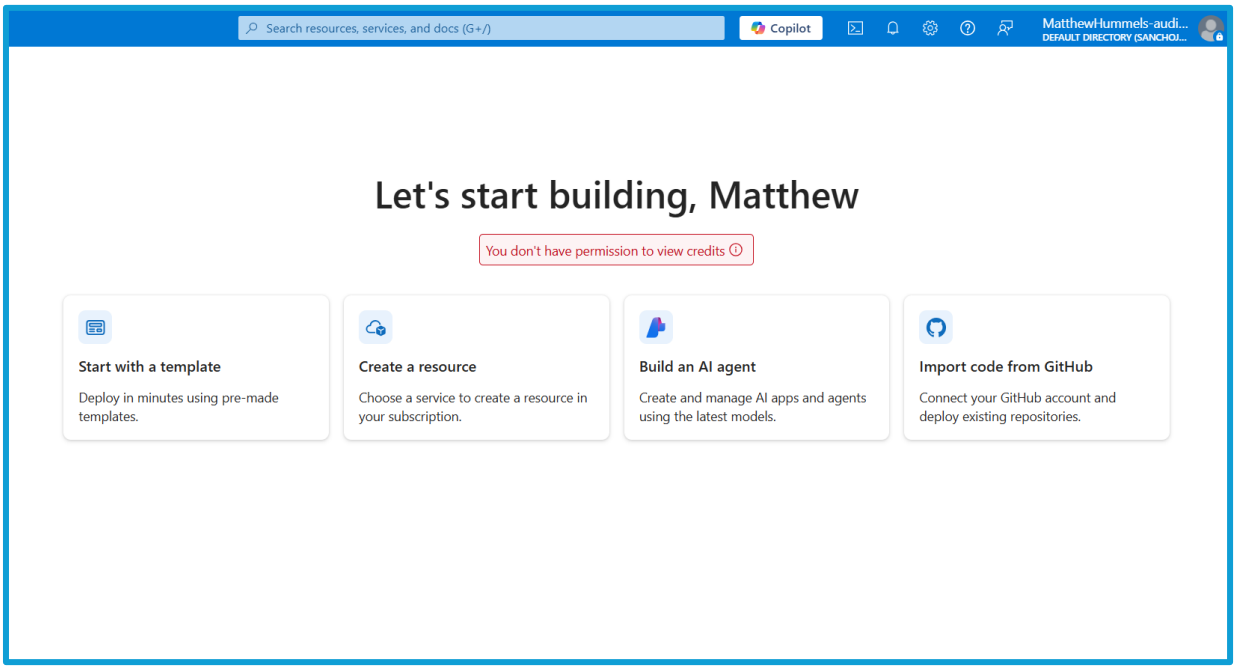

Figure 9: Logged into the second user's profile

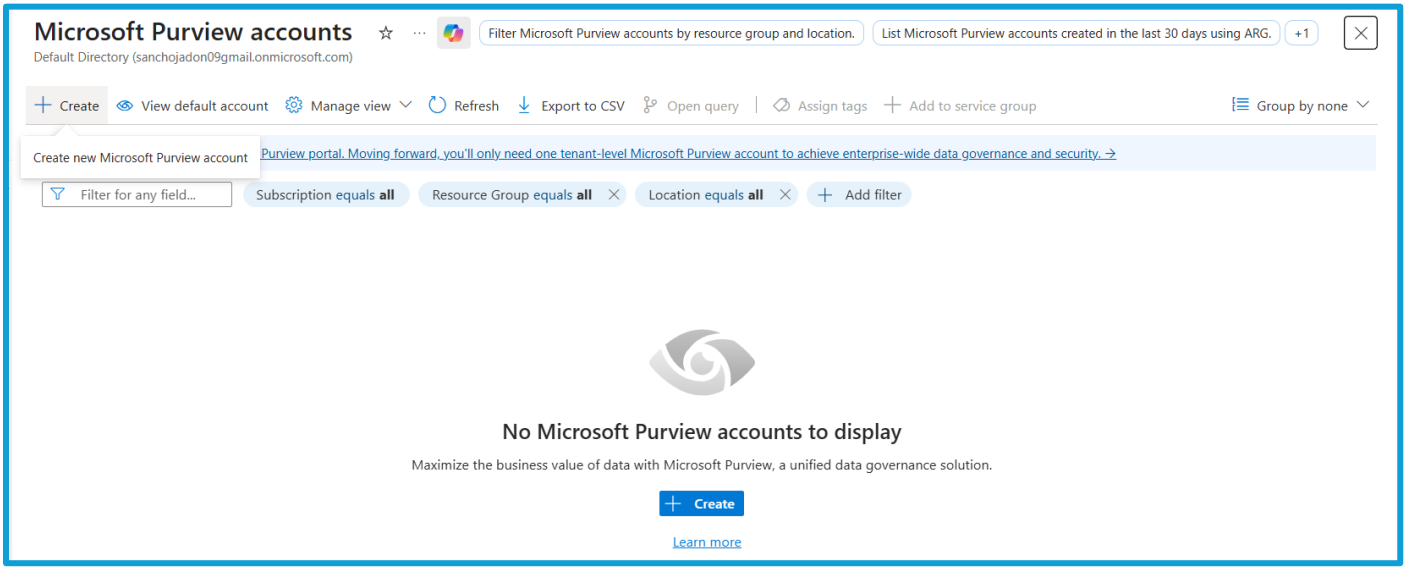

Figure 10: Attempting to create a Purview account with the reader profile

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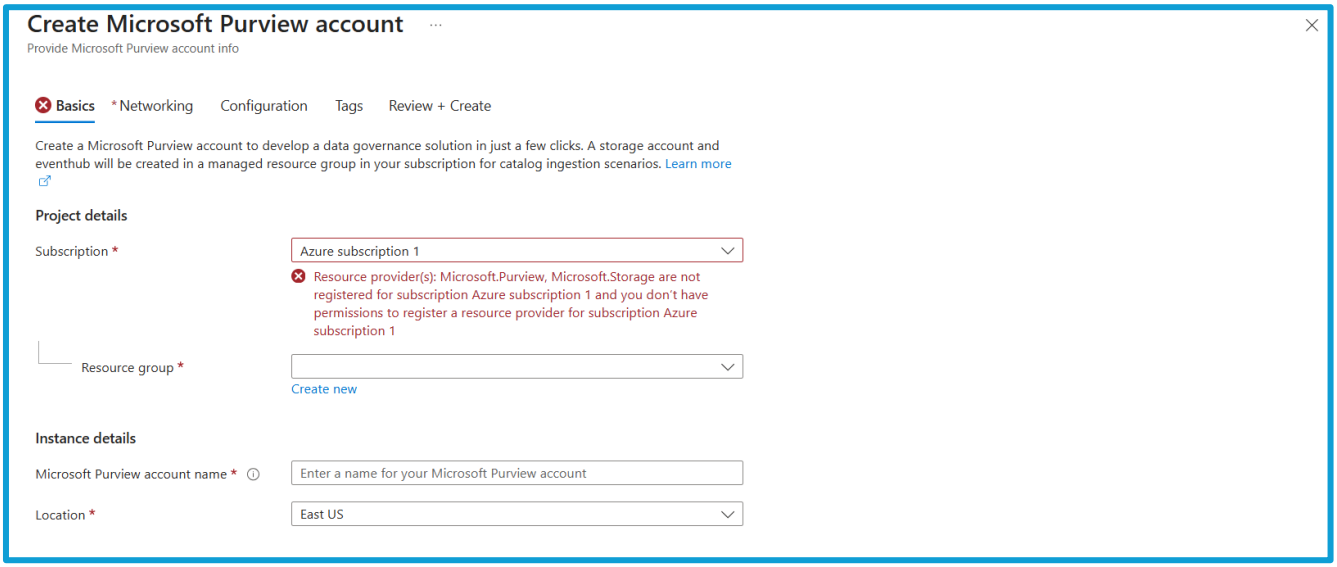

Figure 11: Proof that role assignment was applied to this user

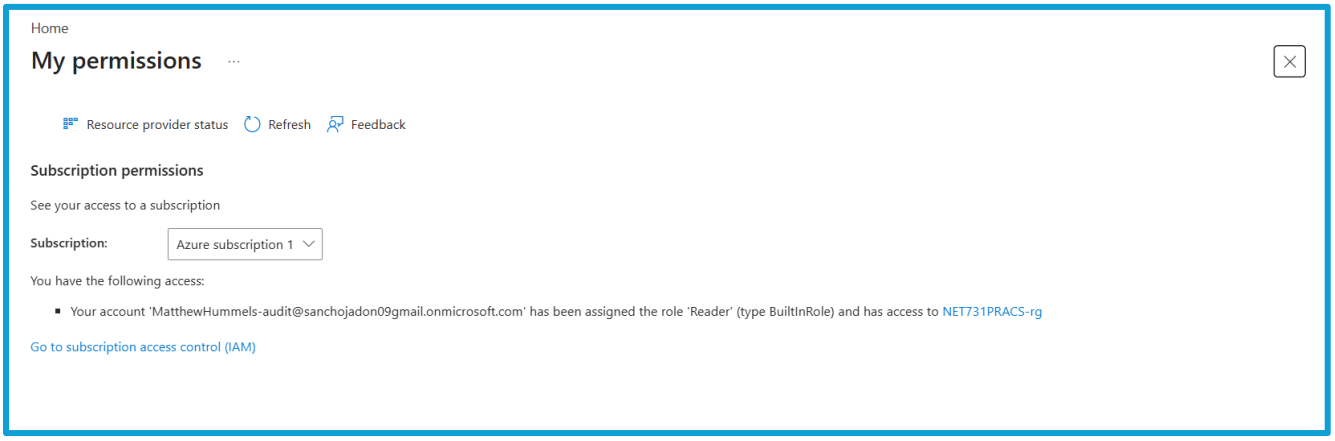

Figure 12: Proof of this user's access

Task 2 — Network Security Group (3 marks)

For the first inbound rule, I chose the TCP protocol along with port 22 because TCP is ideal for remote management of a virtual machine and I chose port 22 specifically because the virtual machine I deployed in week 3 has a Linux image and port 22 is the default port of SSH (Secure Shell). The source and the destination, I entered the ip address (192.168.1.4) and subnet of the virtual machine (192.168.1.0). The second inbound rule is chose the TCP protocol and port 344 which allows internet access traffic. The last outbound rule, the protocol I chose is TCP.

Another explicit deny rule is added to the outbound rule to specifically deny traffic before it reaches a certain point. The priority number I set on this rule is 100 because how priority works is that the lowest number is the highest priority which means lower numbers are evaluated first.

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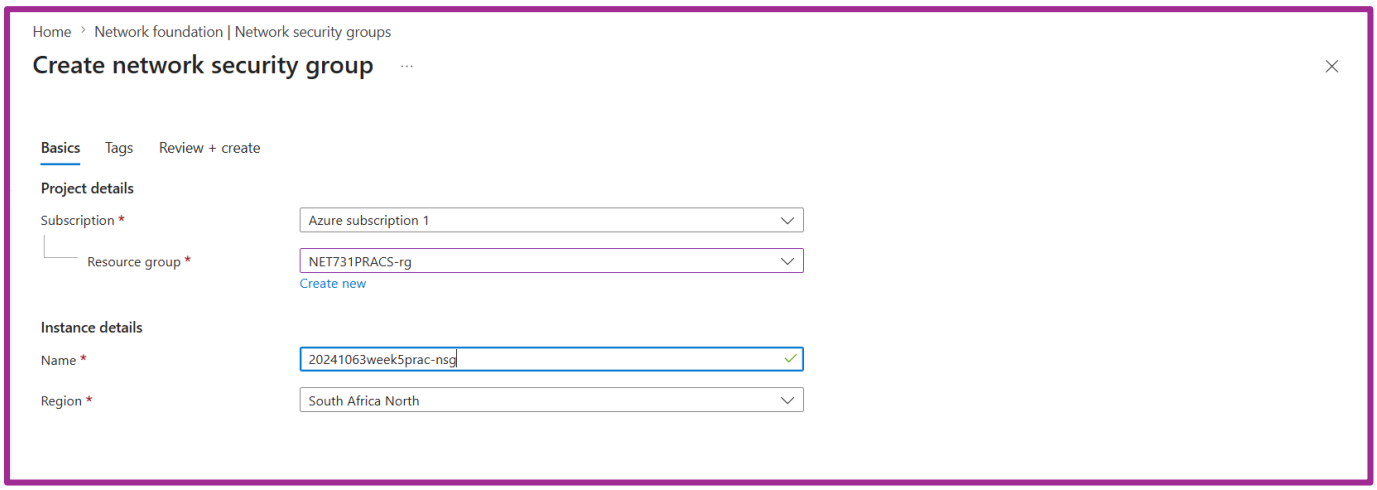

Figure 13: Creating an NSG

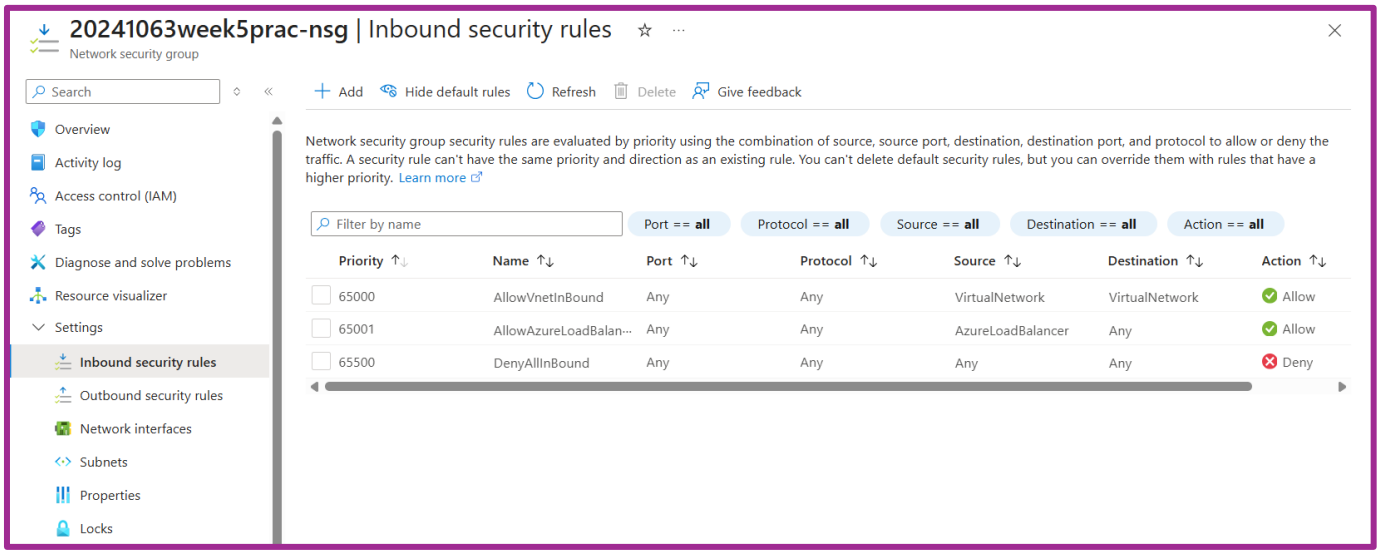

Figure 14: adding the inbound rules on the new NSG

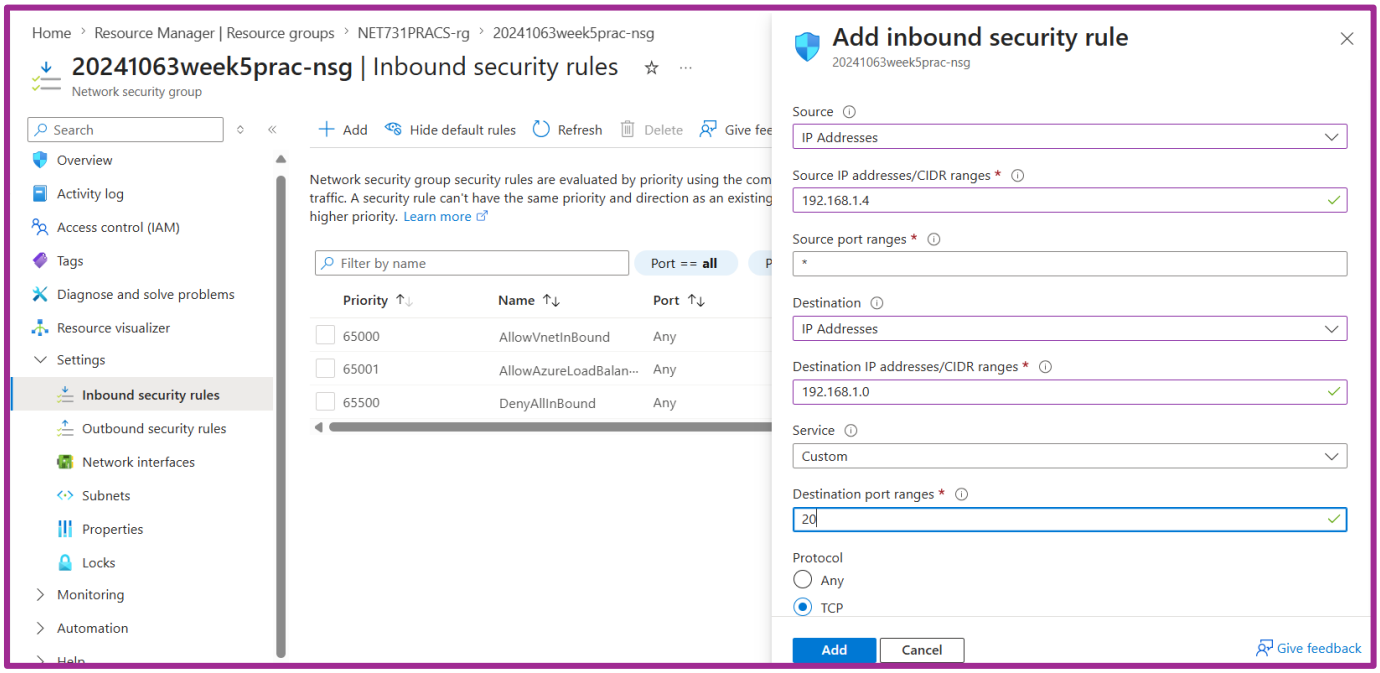

Figure 15: Adding the first NSG inbound rule for remote management of VM (1)

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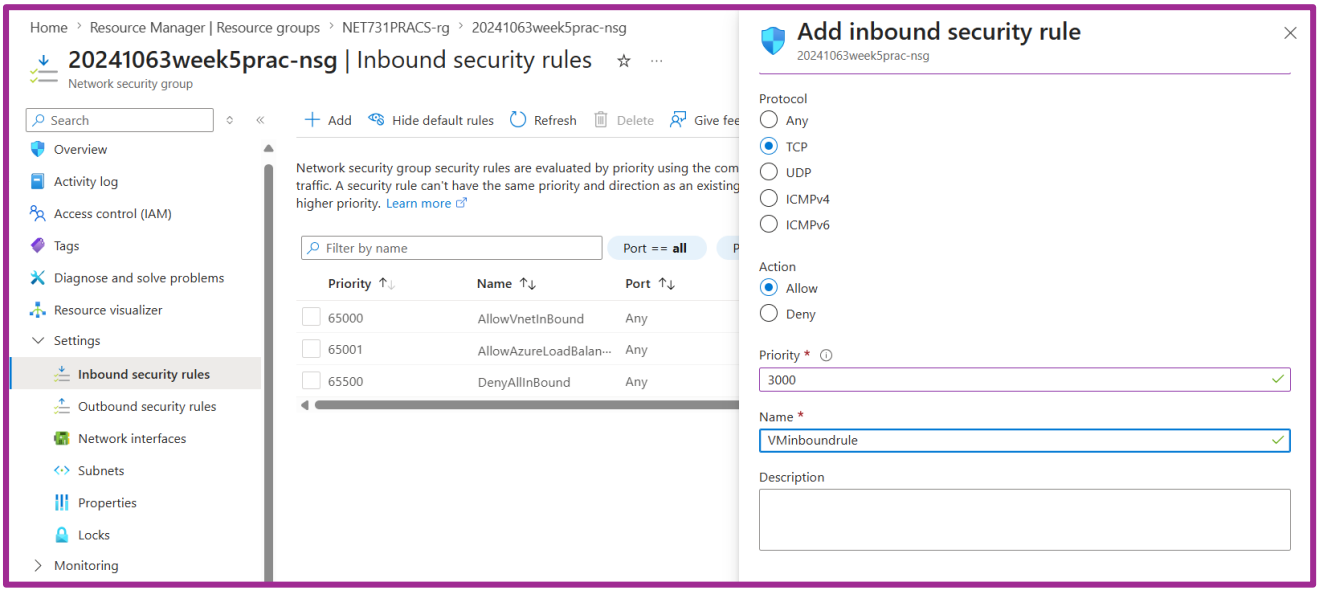

Figure 16: Adding the first NSG inbound rule for remote management of VM (2)

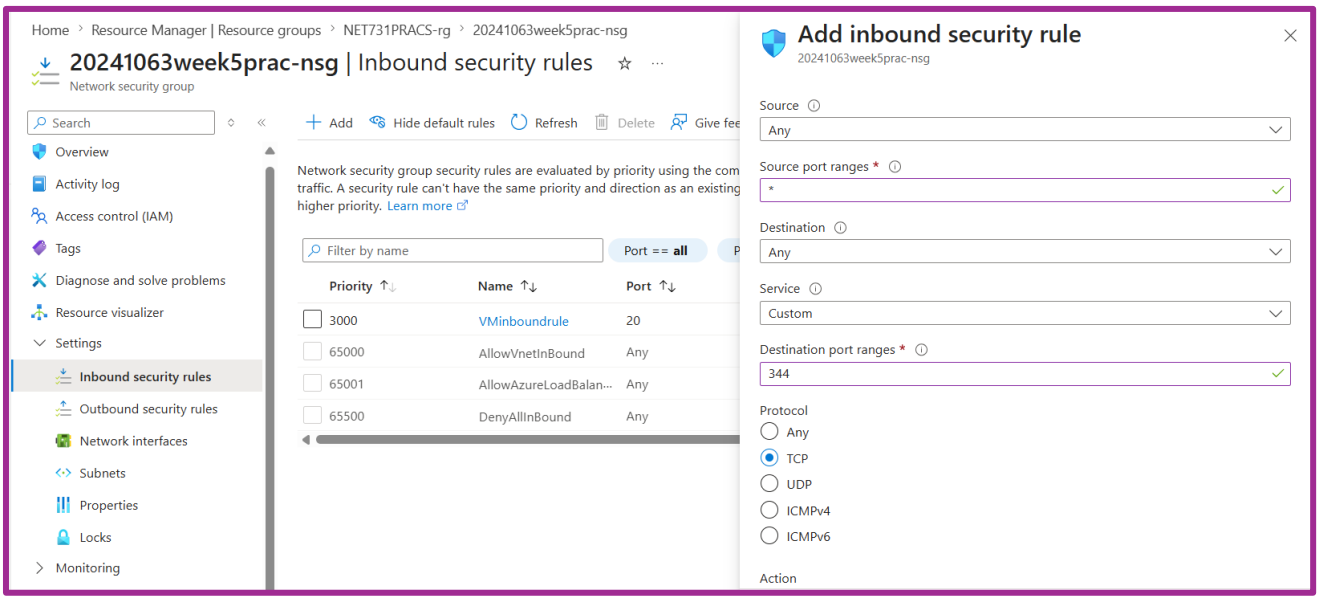

Figure 17: Adding the second inbound NSG rule for web application access (1)

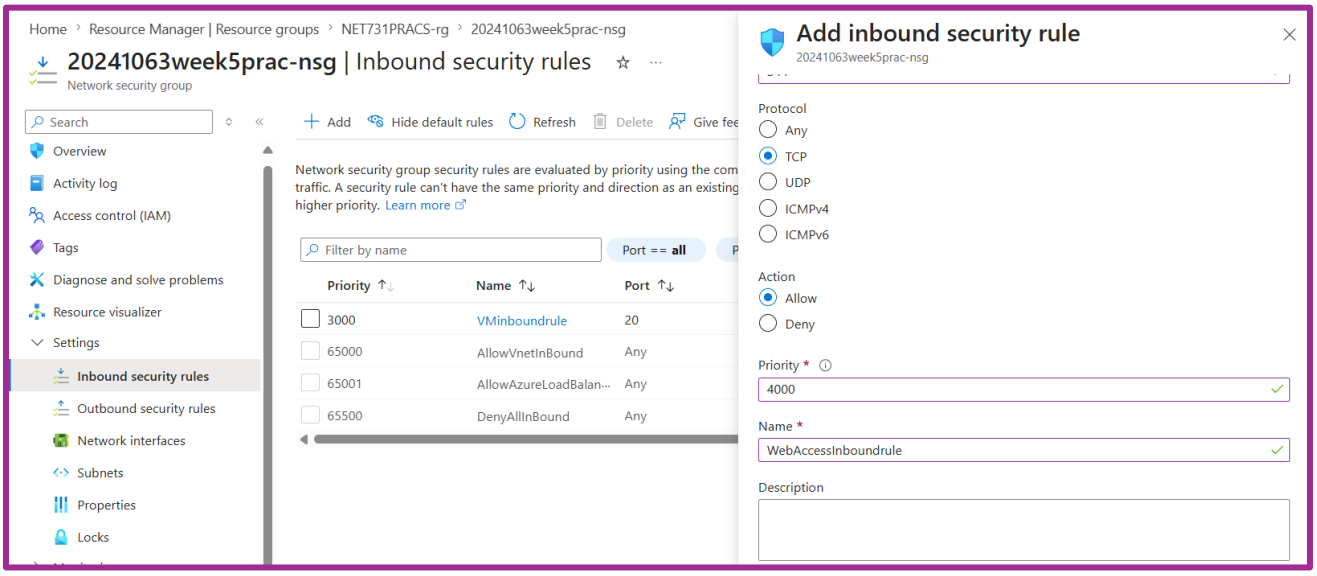

Figure 18: Adding Second inbound NSG rule for web application access (2)

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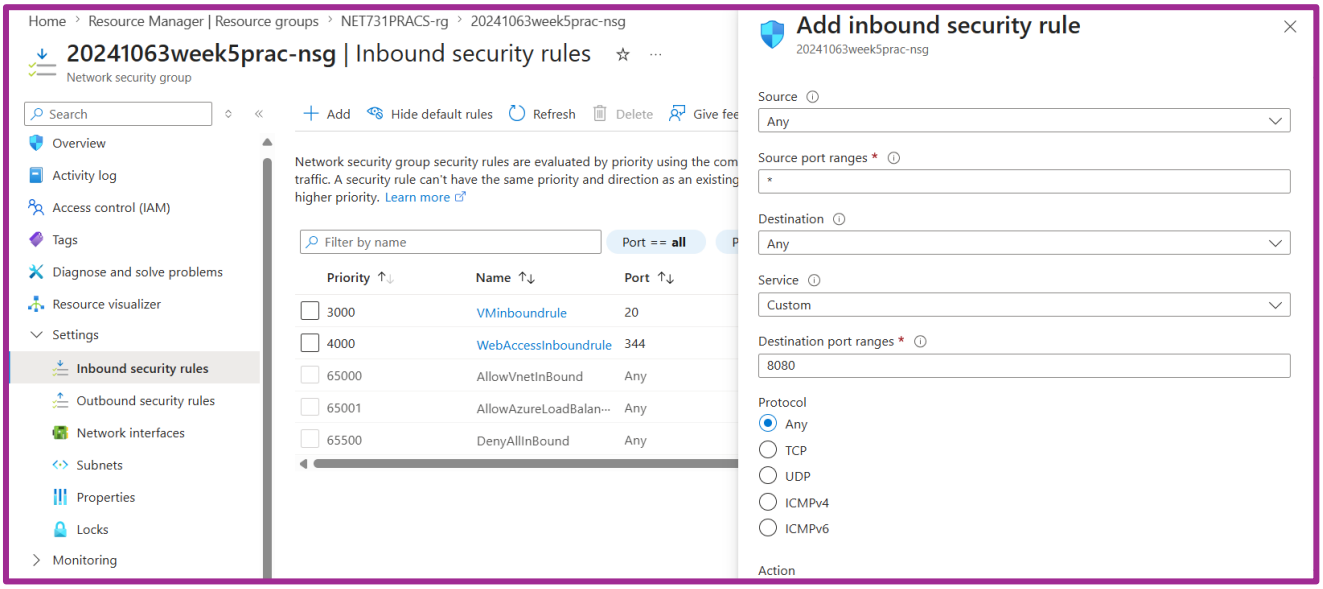

Figure 19: Adding the third inbound rule for least privilege (1)

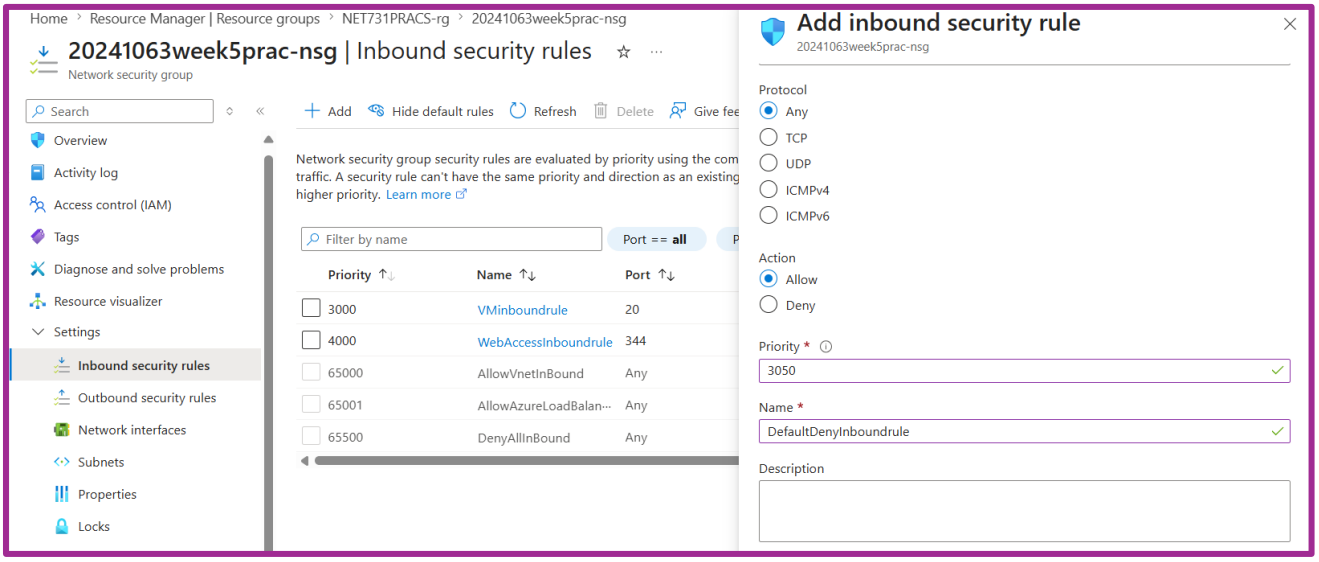

Figure 20: Adding the third inbound rule for least privilege (2)

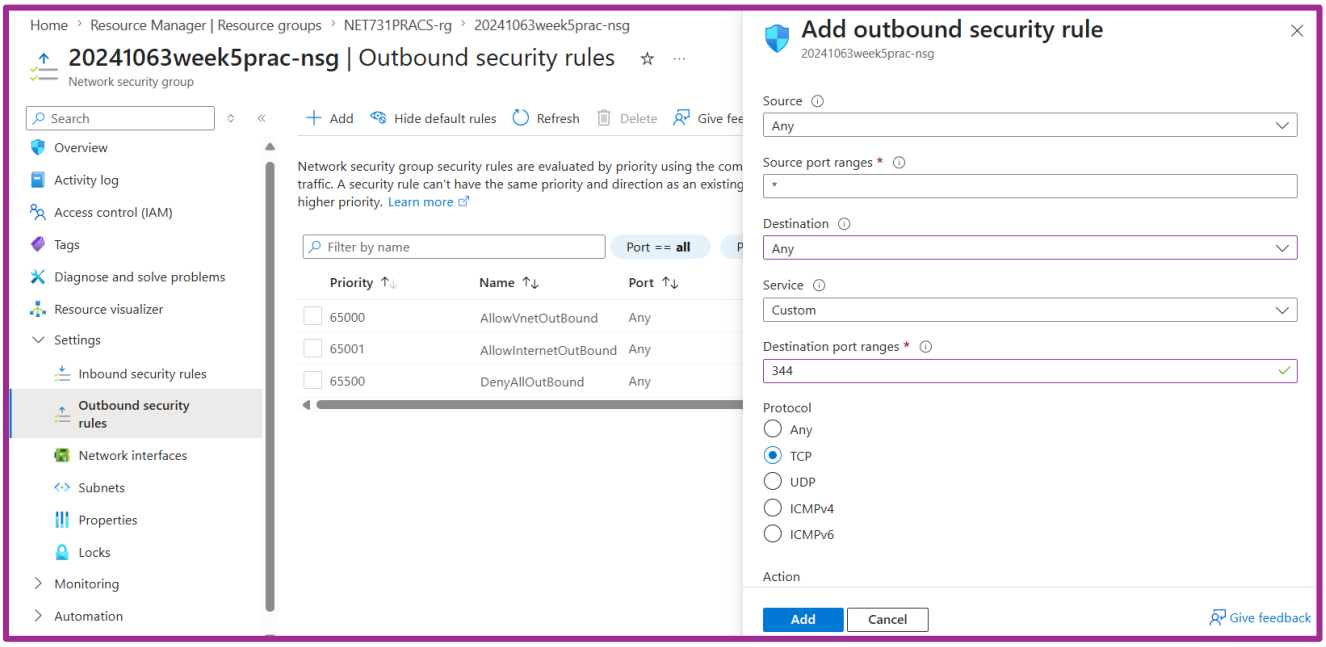

Figure 21: Adding the first outbound rule for data exfiltration prevention for VMs (1)

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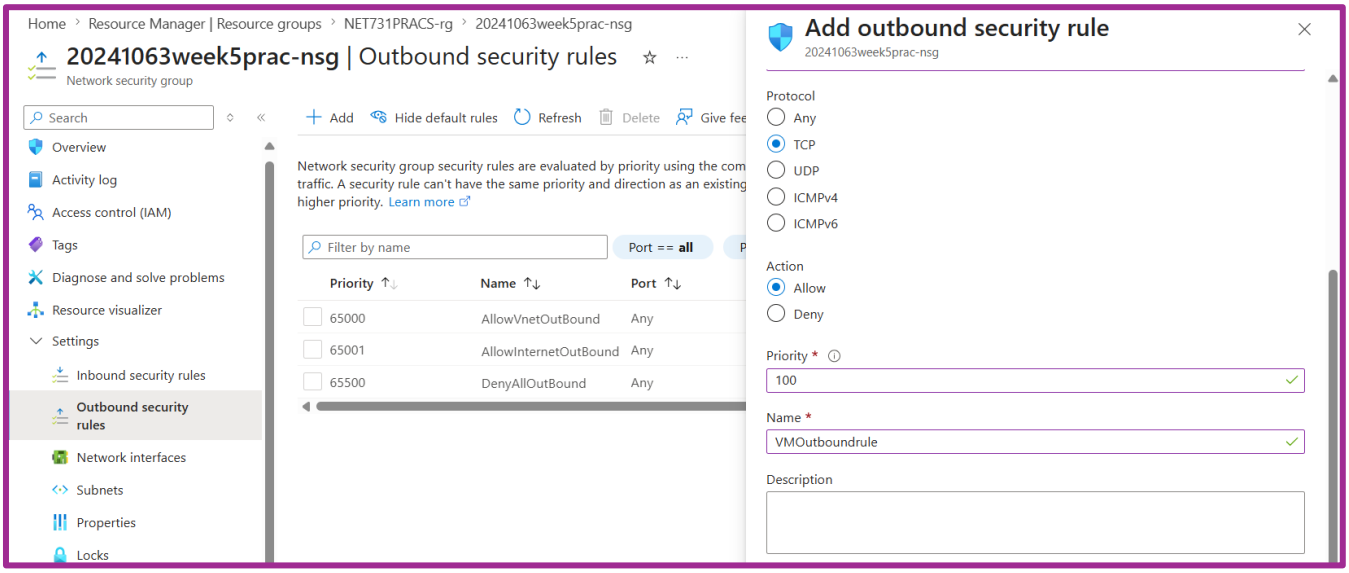

Figure 22: Adding the first outbound rule for data exfiltration prevention for VMs (2)

Task 3 — Storage Account & Containers (2 marks)

For the storage account Redundancy, I chose local redundant storage due to the requirements in the scenario being that it is a small repository with no disaster recovery needed.

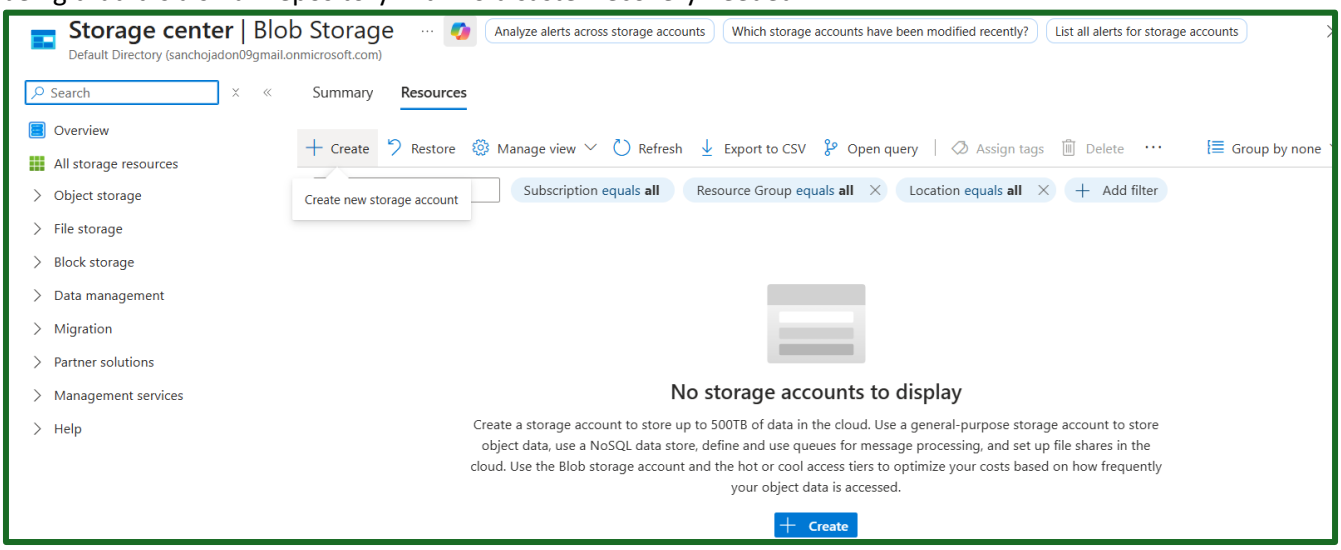

Figure 23: Creating a new storage account

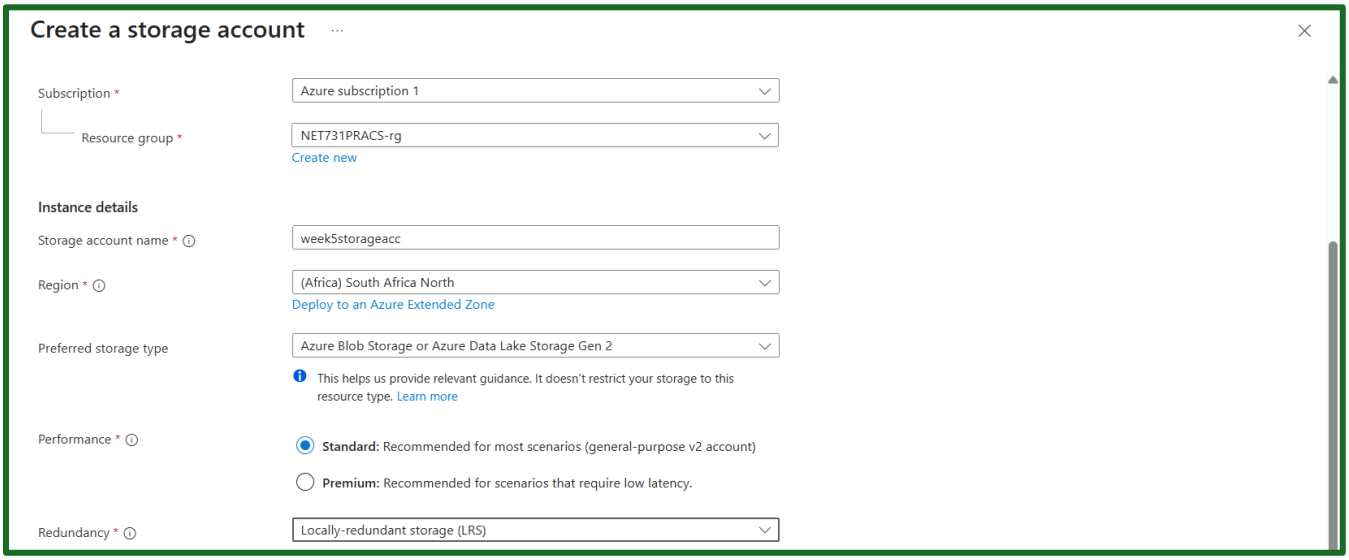

Figure 24: Entering the desired specifications for the new storage account

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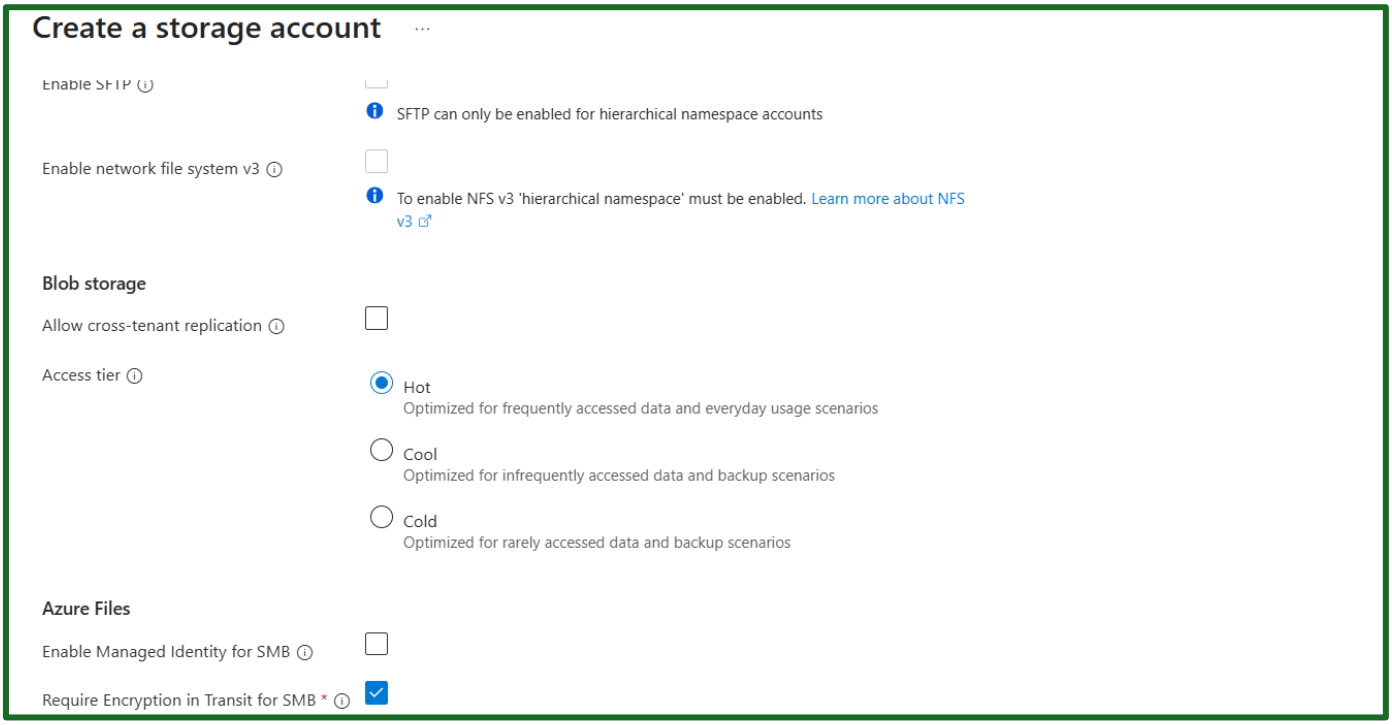

Figure 25: Selecting the access tier for the storage account

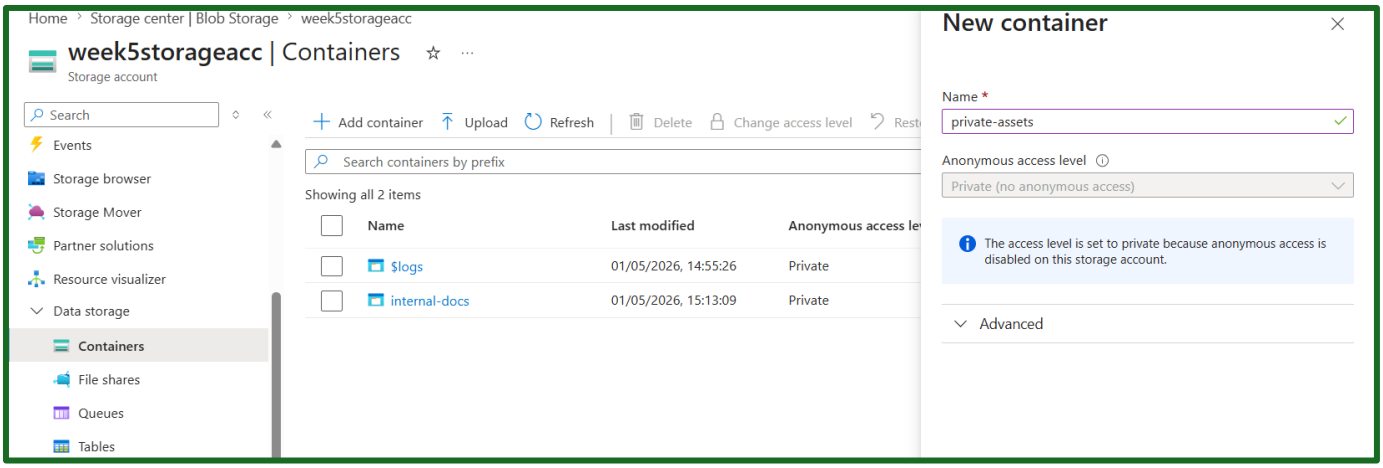

Figure 26: Creating the first container

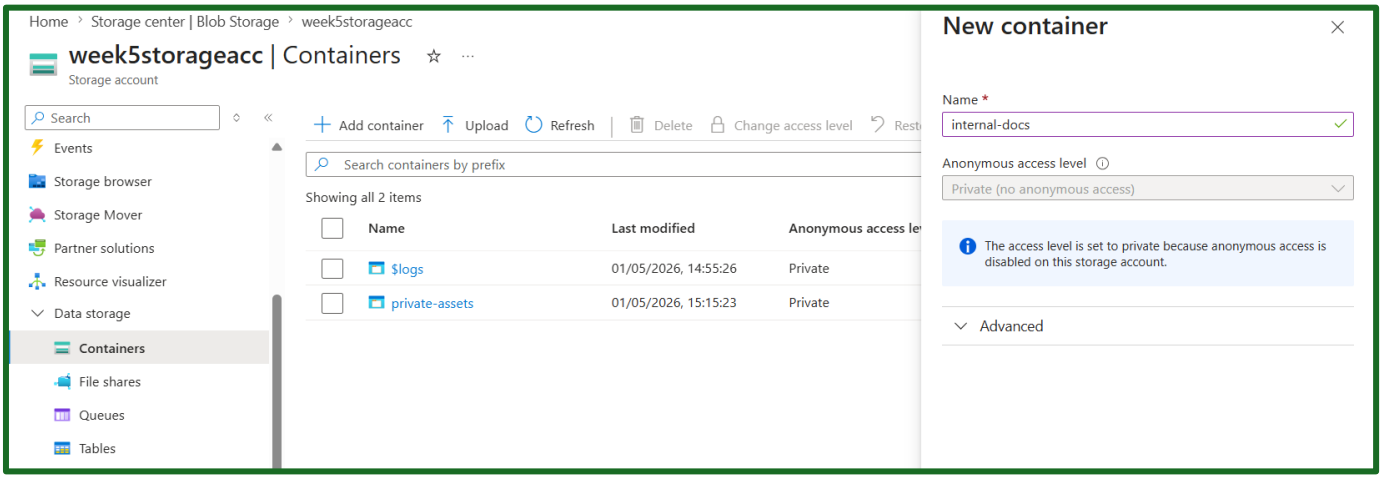

Figure 27: Creating the second container

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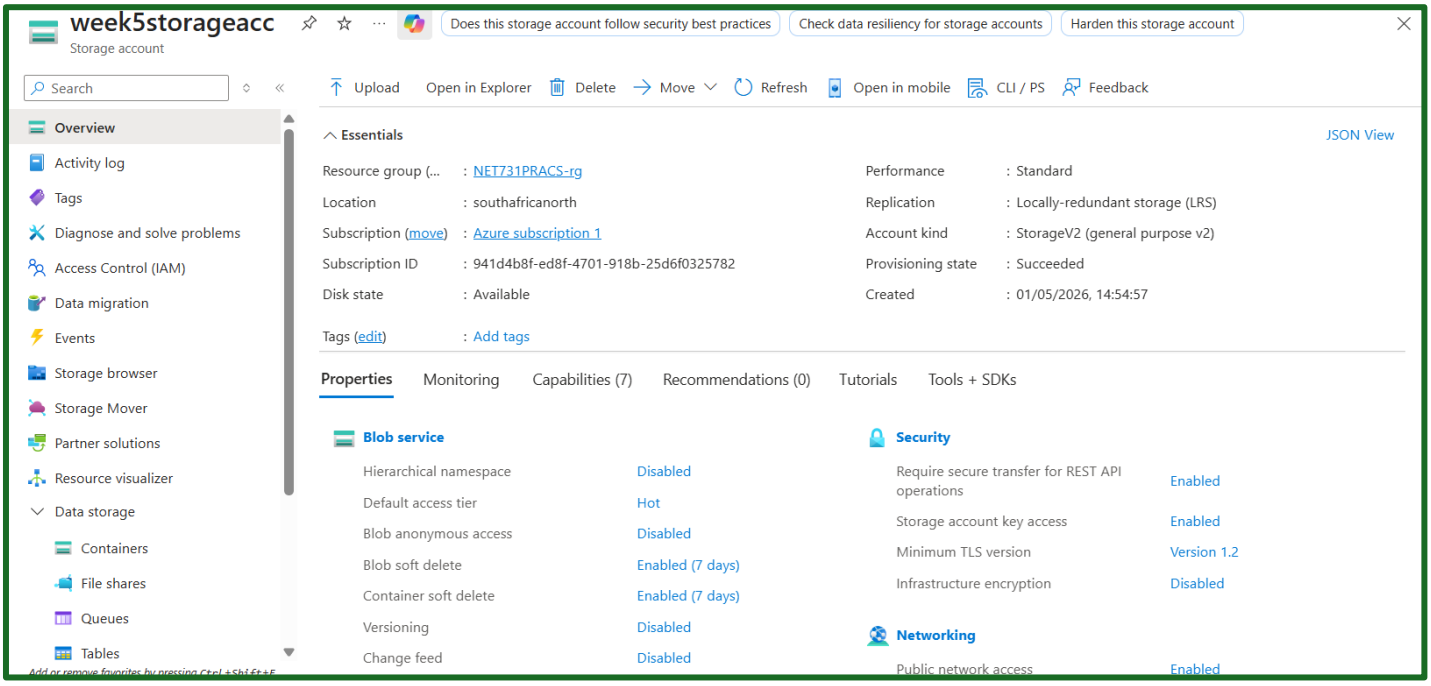

Figure 28: Showing the specifications of the storage account created

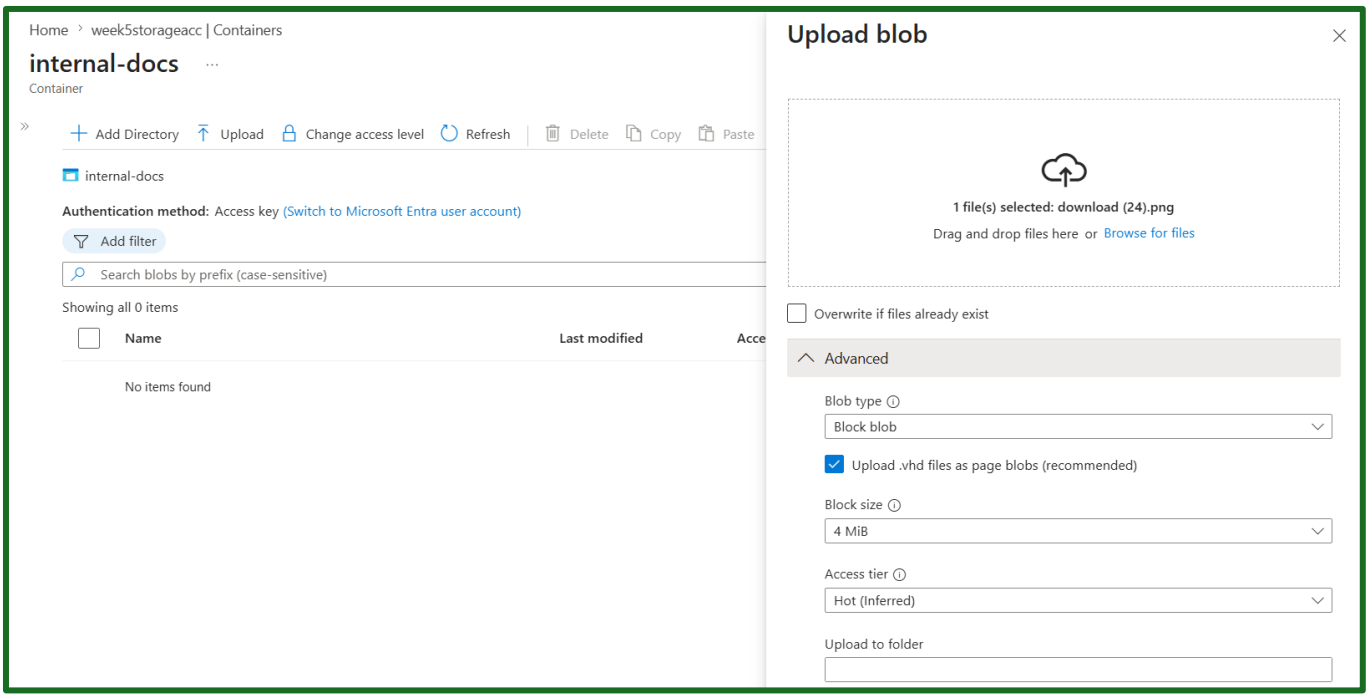

Figure 29: uploading an image in the first container

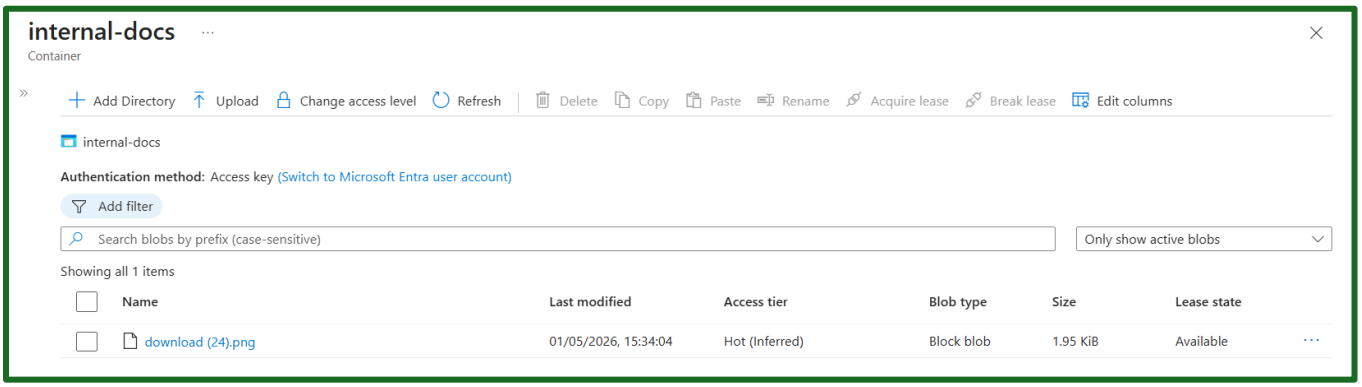

Figure 30: proof of successful image upload

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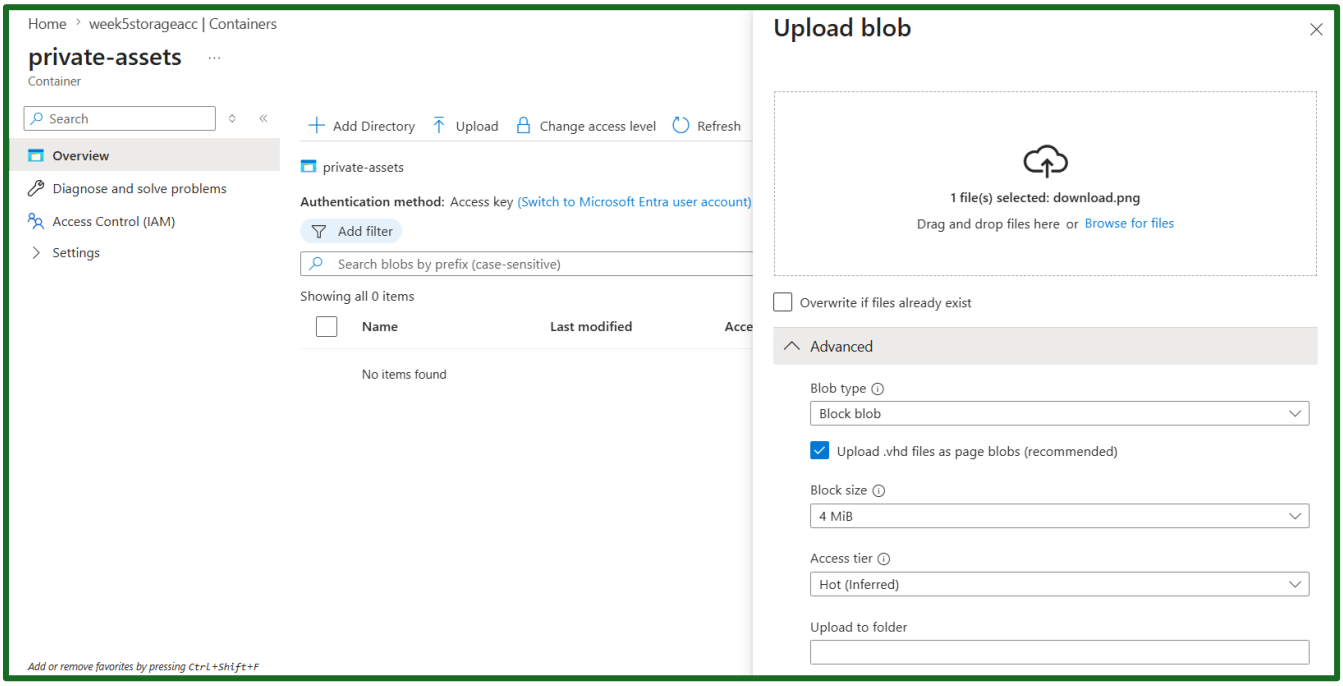

Figure 31: Uploading an image in the second container

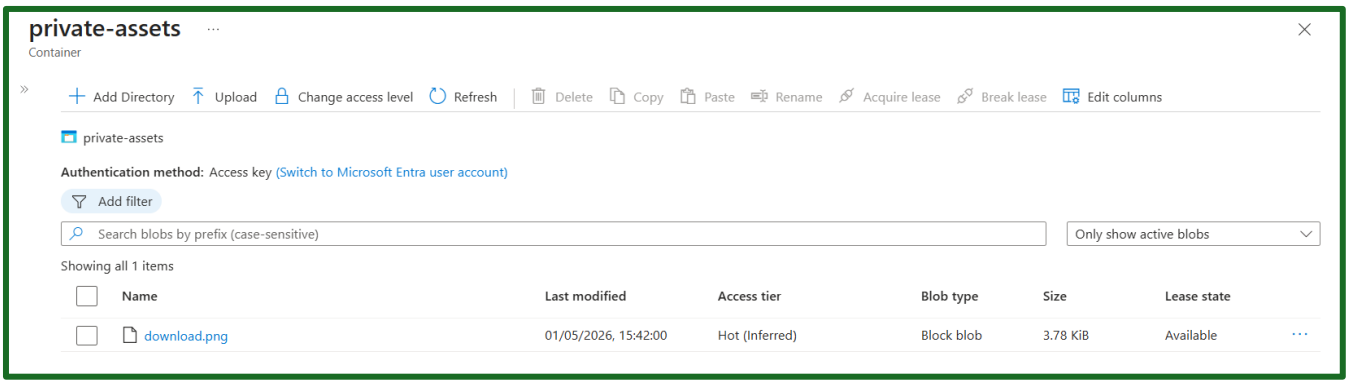

Figure 32: Proof of successful image upload

Task 4 — Storage Access Permissions (2 marks)

For the storage account Redundancy, I chose local redundant storage due to the requirements in the scenario being that it is a small repository with no disaster recovery

Since email is not a very secure medium, sharing extremely sensitive data such as a SAS token provides malicious users a higher chance of getting access to that information. If a SAS token gets lost or stolen, to protect the azure environment then you should regenerate the lost key to prevent unwanted access from third party malware.

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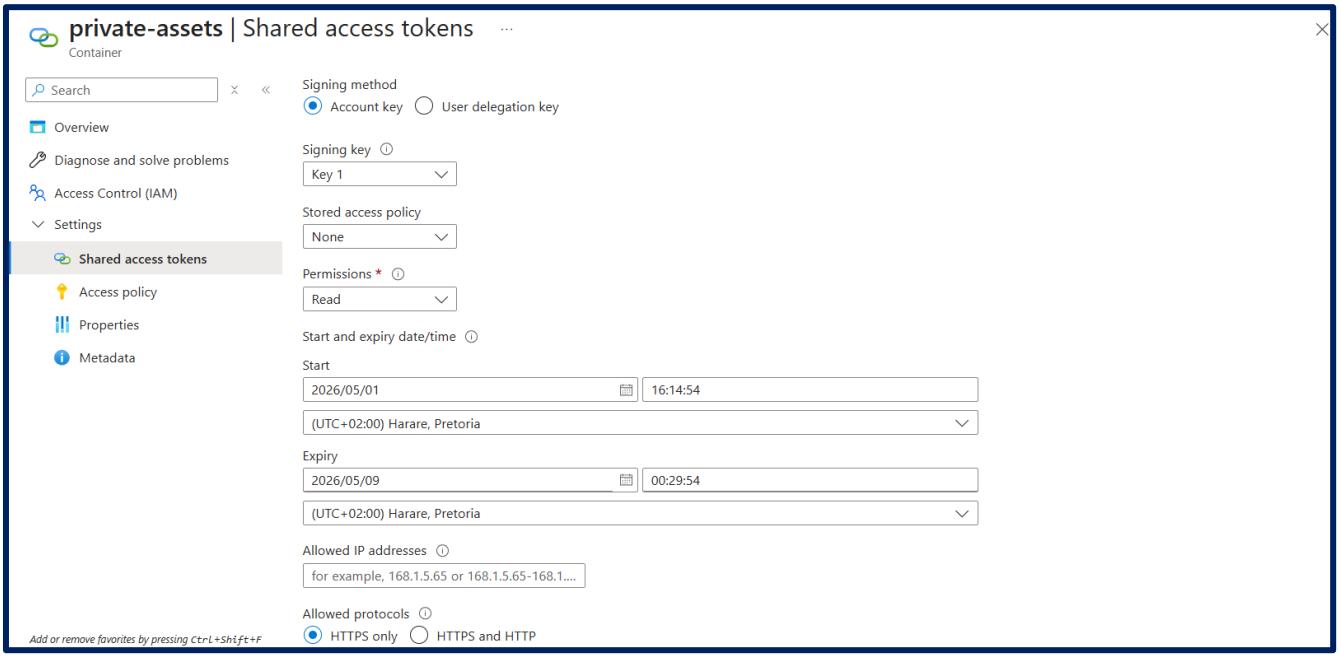
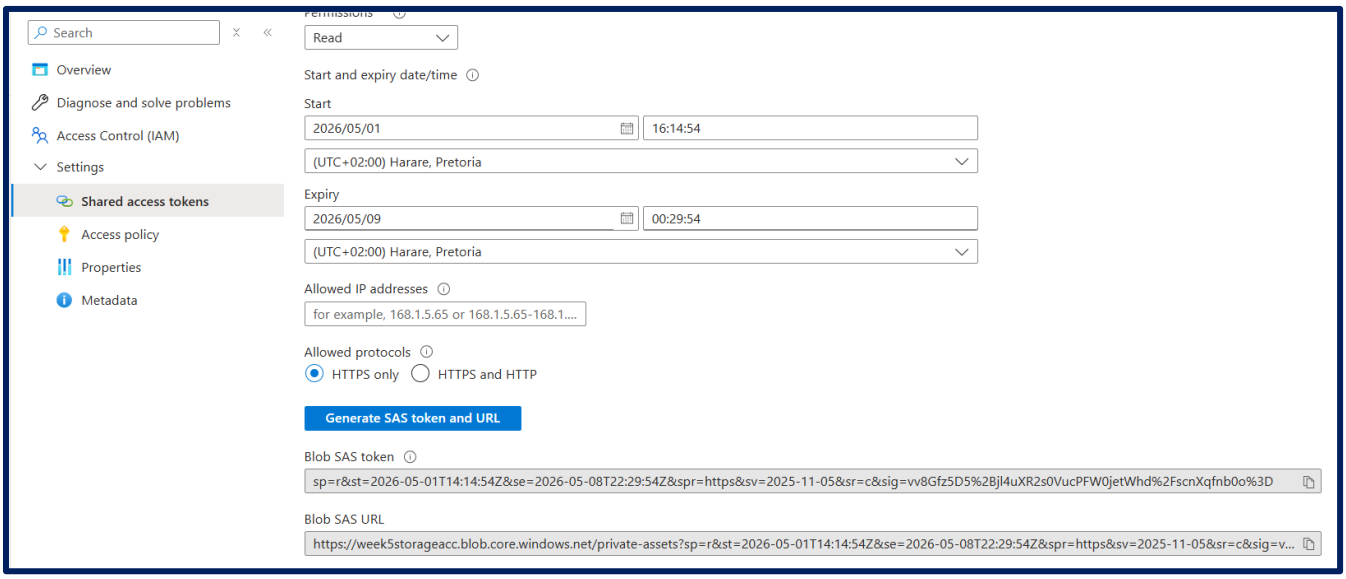


Figure 33: configuring the shared access token



6. Mark Allocation

Criterion	Marks	Awarded	Comment
RBAC role assigned correctly — right role, right scope, right user	3		
NSG inbound and outbound rules configured with clear justification	3		
Storage account created with container and file uploaded successfully	2		
Access permissions configured and evidenced (Blob access level or SAS)	2		
TOTAL	**10**		